A Comparative Analysis of StandardScaler and MinMaxScaler Normalization on Dengue Fever Classification for Deep Neural Network Model

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***Abstract*—Dengue Fever remains one of the major infectious diseases in tropical and subtropical regions, requiring accurate and early diagnosis to reduce complications and mortality. This study proposes a Deep Neural Network (DNN)-based classification framework for Dengue Fever prediction using clinical symptoms and laboratory examination data, with a primary focus on analyzing the impact of normalization techniques on classification performance. The clinical dataset used in this study contains heterogeneous attributes with significantly different numerical ranges, such as platelet count, leukocyte count, hematocrit, hemoglobin, body temperature, and blood pressure, making it challenging to be processed directly by Deep Learning models. Variations in feature scales may negatively affect model stability and classification accuracy, making normalization an important preprocessing step. Three normalization methods, namely no normalization, StandardScaler, and MinMaxScaler, were evaluated using two dataset configurations: a complete dataset and a symptoms-only dataset. Furthermore, 5-Fold Cross Validation was implemented to assess model stability and generalization capability. Experimental results demonstrate that normalization significantly improved classification performance compared to non-normalized data. Among all evaluated methods, StandardScaler achieved the best performance, obtaining an accuracy of 96.96% and an AUC value of 0.9972 on the complete dataset. These findings confirm that normalization plays a crucial role in improving model convergence, training stability, and classification performance for heterogeneous clinical datasets, while the proposed framework shows promising potential for supporting early Dengue Fever prediction and intelligent healthcare decision support systems.**

***Index Terms*—Deep Neural Network, Dengue Fever, Classification, Normalization, StandardScaler, MinMaxScaler, Cross Validation**

I. Introduction

Dengue Hemorrhagic Fever (DHF) remains one of the most prevalent infectious diseases in tropical and subtropical regions worldwide. Transmitted primarily through infected *Aedes aegypti* mosquitoes, dengue continues to pose a major public health challenge, particularly in developing countries with limited healthcare resources. According to the *World Health Organization* (WHO), dengue cases have increased significantly over the past two decades, with an estimated 100–400 million infections annually, especially in Southeast Asia and Latin America [1]. Therefore, timely and accurate diagnosis is essential to reduce complications and mortality associated with DHF [2], [3].

Dengue diagnosis is commonly performed using clinical symptoms and laboratory examinations, including platelet count, leukocyte count, hematocrit, and hemoglobin levels [4]. However, overlapping symptoms with other febrile diseases and the increasing volume of patient data often complicate the diagnostic process, potentially affecting the consistency and accuracy of clinical decisions. In recent years, *Artificial Intelligence* (AI), particularly *Machine Learning* (ML) and *Deep Learning* (DL), has shown promising performance in medical classification and disease prediction due to its ability to learn complex patterns from healthcare data [5], [6].

Among various DL approaches, *Deep Neural Network* (DNN) models have demonstrated strong performance in structured medical classification tasks [7]. Previous studies have successfully applied DNN and ML-based methods for dengue prediction using clinical and laboratory features [8], [9]. Nevertheless, the performance of Deep Learning models is highly dependent on data preprocessing quality, particularly feature normalization. Medical datasets often contain heterogeneous attributes with substantially different numerical ranges, such as platelet count, leukocyte count, temperature, and blood pressure, which may negatively affect model convergence and classification performance if processed directly [10].

Although normalization techniques have been applied in previous dengue classification studies, comparative analysis of their impact on DNN performance remains limited. Therefore, this study investigates the effect of different normalization methods on a DNN-based Dengue Fever classification model using *complete* and *symptoms-only* datasets. Three approaches, namely without normalization, *StandardScaler*, and *MinMaxScaler*, are evaluated, while *K-Fold Cross Validation* is implemented to assess model stability and generalization capability [11]. Model performance is evaluated using Accuracy, Precision, Recall, F1-Score, and ROC-AUC metrics to support more reliable early Dengue Fever prediction.

II. Related Work

Previous studies have demonstrated that *Machine Learning* (ML) and *Deep Learning* (DL) approaches are effective for Dengue Fever classification and prediction. A systematic review conducted by Andrade Girón et al. [12] reported that various models, including *Artificial Neural Network* (ANN), *Convolutional Neural Network* (CNN), *Random Forest*, *Support Vector Machine* (SVM), and *XGBoost*, have been widely applied for dengue diagnosis. The study also emphasized that preprocessing techniques, particularly normalization and *K-Fold Cross Validation*, play an important role in improving model performance and generalization.

Several studies have highlighted the importance of normalization and validation strategies in healthcare classification systems. Previous findings reported that normalization methods, such as *Z-score normalization*, improve Deep Learning performance on biomedical datasets, while *5-Fold* and *10-Fold Cross Validation* help reduce overfitting and improve model reliability [13]–[15]. In addition, normalization techniques contribute to stabilizing the training process and improving model convergence.

In dengue classification research, both ML and DL methods have achieved promising performance using normalized clinical datasets. Hamdani et al. [16], [17] applied *Support Vector Machine* (SVM) and *Artificial Neural Network* (ANN) models with feature normalization and *K-Fold Cross Validation*, achieving accuracies above 97%. Similarly, Ramasamy et al. [18], [19] proposed a Neural Network model integrated with *StandardScaler* normalization and feature optimization, achieving 97.1% accuracy using *5-Fold Cross Validation*. Other studies also reported that normalization techniques, including *MinMaxScaler*, improved prediction reliability and consistency in dengue forecasting systems [20], [21].

Comparative studies involving multiple ML algorithms have also shown strong classification performance for dengue prediction. Abdualgalil et al. [3] employed *Z-score normalization*, *SMOTE+ENN* balancing, and *10-Fold Cross Validation*, where the *Extra Tree Classifier* achieved 99.03% accuracy. Likewise, Sriyanto et al. [22] reported that *Neural Network* and *Decision Tree* models achieved the highest accuracy of 99.4% using *StandardScaler* normalization. In the Indonesian context, Dewi et al. [5] demonstrated that *XGBoost* achieved the best performance with 85% accuracy after applying *Min-Max scaling*, while Xu and Xiao [23] emphasized the importance of feature scaling and validation techniques in improving model generalization.

Although previous studies have demonstrated the benefits of normalization and *Cross Validation* in dengue prediction, comparative analysis of multiple normalization methods under different dataset configurations for DNN-based Dengue Fever classification remains limited. Therefore, this study focuses on analyzing the impact of different normalization techniques and validation scenarios using both *complete clinical-laboratory datasets* and *symptoms-only datasets*.

III. Research Methodology

A. Research Workflow

The research workflow was designed to evaluate the impact of normalization methods on the proposed Deep Neural Network (DNN) model for Dengue Fever classification, including preprocessing, normalization, model training, Cross Validation, and performance evaluation.

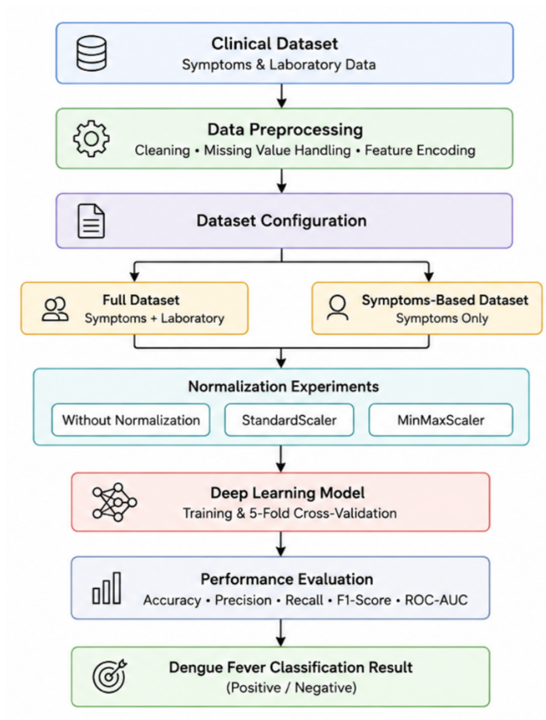

Fig. 1. Research Workflow

Fig. 1 illustrates the overall workflow used in this study. The process begins with collecting clinical symptom and laboratory examination datasets related to Dengue Fever. The collected data then undergoes preprocessing stages, including data cleaning, missing value handling, feature transformation, and normalization. After preprocessing, the dataset was divided into two dataset configurations, namely the complete dataset containing symptom and laboratory features and the symptoms-only dataset.

Furthermore, normalization experiments were conducted using three approaches: without normalization, *StandardScaler*, and *MinMaxScaler*. Each experimental configuration was evaluated using a Deep Neural Network (DNN) model combined with 5-Fold Cross Validation to analyze model stability and generalization capability. Finally, model performance was evaluated using several classification metrics, including Accuracy, Precision, Recall, F1-Score, and ROC-AUC, to determine the most effective normalization approach for Dengue Fever classification.

B. Dataset

The dataset used in this study consists of clinical data related to Dengue Fever (DF) patients, including demographic information, vital signs, laboratory examination results, and clinical symptoms. The dataset was collected to support the development of a Deep Neural Network (DNN)-based classification system for Dengue Fever prediction.

Overall, the dataset contains 20 input features and 1 target label. The input features are categorized into demographic data, vital signs, laboratory examination features, and clinical symptoms. Meanwhile, the target label represents the diagnosis result used for the classification process.

TABLE I
INPUT FEATURE CATEGORIES

Category	Attributes	Exp Applied
Demographic Data	Age, Gender	Age Only
Vital Signs	Temperature, Systolic Blood Pressure, Diastolic Blood Pressure	Yes
Laboratory Features	Platelet, Leukocyte, Hematocrit, Hemoglobin	Yes
Clinical Symptoms	Fever, Fever Duration, Headache, Nausea, Vomiting, Abdominal Pain, Fatigue, Loss of Appetite, Red Spots, Body Pain, Abdominal Distension, Joint Pain	Fever Duration Only

TABLE II
TARGET LABEL DESCRIPTION

Target Label	Description
0	Fever
1	Positive Dengue Fever

Normalization was applied only to numerical attributes with continuous values, including age, body temperature, blood pressure, laboratory examination results, and fever duration. Binary and categorical symptom features that had been transformed into numerical values (0 and 1) were not normalized because they already shared the same value range. The target label was also excluded from the normalization process since it represents the classification output of the proposed DNN model.

C. Dataset Scenarios

This study used two dataset configurations to examine the impact of dataset composition on Deep Learning-based Dengue Fever classification. The first configuration employed a complete dataset consisting of demographic information, vital signs, laboratory examination results, and clinical symptoms. In contrast, the second configuration included only symptom-related features to simulate early Dengue Fever prediction in situations where laboratory test results are unavailable.

TABLE III
DATASET CONFIGURATIONS

Dataset Configuration	Dataset Composition
Dataset 1	Demographic Data + Vital Signs + Clinical Symptoms + Laboratory Features
Dataset 2	Demographic Data + Vital Signs + Clinical Symptoms

D. Data Preprocessing

Data preprocessing was conducted to prepare the dataset before the Deep Learning training process. This stage aims to improve data quality and ensure that all features can be processed effectively by the model.

- **Data Cleaning**
  The dataset was examined to identify duplicated, inconsistent, and incomplete records. Duplicate data were removed to reduce bias and improve model reliability.
- **Missing Value Handling**
  Missing values in numerical attributes such as temperature, platelet count, leukocyte count, hematocrit, and hemoglobin were handled using mean imputation. Meanwhile, categorical attributes were filled using mode imputation to maintain dataset consistency.
- **Data Transformation**
  Categorical attributes such as symptoms, gender, and diagnosis labels were transformed into numerical values using binary encoding. Positive responses were represented as 1, while negative responses were represented as 0.
- **Data Normalization**
  Feature normalization was applied to reduce differences in numerical ranges among attributes. In this study, three normalization methods were evaluated, namely without normalization, *StandardScaler*, and *MinMaxScaler*.
- **Feature Selection**
  Feature selection was performed based on medical relevance and correlation analysis. All selected features were retained because they showed significant contribution to Dengue Fever classification.
- **Data Splitting**
  The dataset was divided into training and testing data using an 80:20 ratio. The training data were used for model learning, while the testing data were used for performance evaluation.

E. Normalization Experiments

Normalization experiments were conducted to evaluate the effect of feature scaling on the performance of the proposed Deep Neural Network (DNN) model. Since the dataset contains numerical attributes with different value ranges, normalization was applied to improve training stability and classification performance.

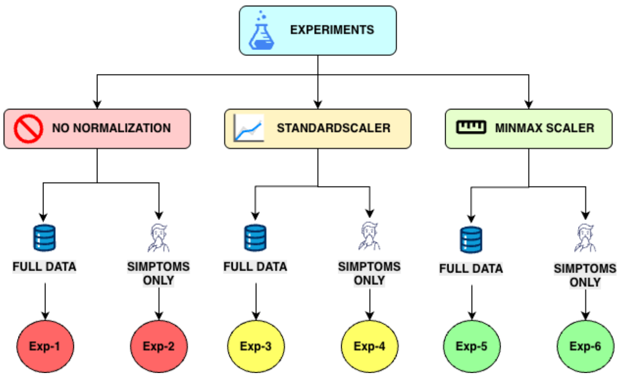

Fig. 2. Normalization Experiment Framework

Fig. 2 illustrates the experimental framework used in this study. Three normalization methods, namely without normalization, *StandardScaler*, and *MinMaxScaler*, were applied to two dataset configurations: the complete dataset and the symptoms-only dataset. Each configuration was evaluated using the proposed Deep Neural Network model to analyze the effect of feature scaling on classification performance.

- **Without Normalization**
  The original dataset was directly used without applying any feature scaling process.
- **StandardScaler**
  StandardScaler transforms the data into a standard normal distribution with a mean of 0 and a standard deviation of 1 [24]. The normalization process is calculated using the following equation:

$$z = \frac{x - \mu}{\sigma} \tag{1}$$

  where x represents the original feature value, μ is the mean value, and σ is the standard deviation.
- **MinMaxScaler**
  MinMaxScaler scales feature values into the range between 0 and 1 [25]. The normalization process is defined as follows:

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}} \tag{2}$$

  where x is the original value, x_{min} is the minimum feature value, and x_{max} is the maximum feature value.

The performance of each experiment was compared using classification evaluation metrics.

TABLE IV
NORMALIZATION EXPERIMENT CONFIGURATIONS

Experiment	Dataset	Normalization
Exp-1	Complete Dataset	None
Exp-2	Complete Dataset	StandardScaler
Exp-3	Complete Dataset	MinMaxScaler
Exp-4	Symptoms Dataset	None
Exp-5	Symptoms Dataset	StandardScaler
Exp-6	Symptoms Dataset	MinMaxScaler

F. Proposed Deep Neural Network Model

Deep Neural Network (DNN) models are widely used for classification tasks because of their capability to learn complex non-linear relationships from structured data [26]. ReLU activation is commonly applied in hidden layers to improve computational efficiency and reduce the vanishing gradient problem [27]. Furthermore, dropout regularization is frequently used to minimize overfitting during the training process [28]. For binary classification problems, the Sigmoid activation function is generally used in the output layer to produce probability values between 0 and 1 [29].

TABLE V
PROPOSED DEEP NEURAL NETWORK ARCHITECTURE

Layer	Configuration	Activation
Input Layer	Depends on dataset configuration	-
Dense Layer 1	32 Neurons	ReLU
Dropout Layer 1	Rate = 0.2	-
Dense Layer 2	16 Neurons	ReLU
Dropout Layer 2	Rate = 0.1	-
Output Layer	1 Neuron	Sigmoid

TABLE VI
TRAINING HYPERPARAMETERS

Parameter	Value
Optimizer	Adam
Loss Function	Binary Crossentropy
Epochs	50
Batch Size	32
Learning Rate	0.001
Early Stopping	Patience = 5
Cross Validation	5-Fold
Decision Threshold	0.4

G. Evaluation Metrics

The proposed Deep Learning model was evaluated using Accuracy, Precision, Recall, F1-Score, and ROC-AUC metrics to measure classification performance and model reliability [30], [31].

- **Accuracy**

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{3}$$

- **Precision**

$$Precision = \frac{TP}{TP + FP} \tag{4}$$

- **Recall**

$$Recall = \frac{TP}{TP + FN} \tag{5}$$

- **F1-Score**

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{6}$$

- **ROC-AUC**

ROC-AUC was used to evaluate the classification capability of the model across different threshold values.

Confusion Matrix analysis was also performed to compare classification results for each normalization method and dataset configuration.

IV. Results

This section presents the experimental results and analysis of the proposed Deep Learning model for Dengue Fever classification. The evaluation focuses on the influence of normalization methods and dataset configurations on classification performance.

A. Model Training Performance

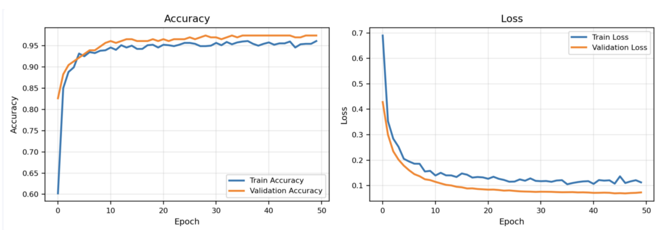

Fig. 3. Training Accuracy and Loss

Fig. 3 shows the training and validation performance of the proposed Deep Learning model. The accuracy graph indicates that both training and validation accuracy increased consistently during the training process and reached stable performance after several epochs. The model achieved validation accuracy above 97%, indicating good classification capability.

Meanwhile, the loss graph demonstrates a gradual decrease in both training and validation loss values. This result indicates that the model successfully minimized classification errors and learned the dataset patterns effectively. In addition, the relatively small gap between training and validation performance suggests that the model achieved good generalization capability with minimal overfitting.

B. Normalization Experiment Results

Normalization experiments were conducted to analyze the effect of feature scaling on the performance of the proposed Deep Neural Network (DNN) model. Six experimental scenarios were evaluated using two dataset configurations, namely the complete dataset and the symptoms-only dataset. Each dataset configuration was tested using three normalization methods: without normalization, *StandardScaler*, and *MinMaxScaler*. In addition, 5-Fold Cross Validation was applied in each experiment to obtain more stable and reliable evaluation results.

Table VII presents an example comparison of feature values before and after applying normalization techniques. The results show that *StandardScaler* transforms the data into a standardized distribution, while *MinMaxScaler* scales feature values into a range between 0 and 1.

TABLE VII
COMPARISON OF DATA BEFORE AND AFTER NORMALIZATION

Attribute	Without Normalization	Standard Scaler	MinMax Scaler
Age (Years)	26	0.03	0.33
Temperature (°C)	36.7	-1.15	0.18
Systolic Blood Pressure	100	0.05	0.40
Diastolic Blood Pressure	70	0.30	0.25
Platelet	139000	-0.33	0.05
Leukocyte	4500	-0.64	0.07
Hematocrit	26.7	-1.63	0.29
Hemoglobin	13.5	-0.30	0.63
Fever Duration (Days)	5	1.43	0.71

TABLE VIII
NORMALIZATION EXPERIMENT AND CROSS VALIDATION RESULTS

Experiment	Accuracy	Precision	Recall	F1-Score
Exp-1	60.87%	69.23%	39.13%	50.00%
Exp-2	**96.96%**	**95.76%**	**98.26%**	**97.00%**
Exp-3	95.22%	93.33%	97.39%	95.32%
Exp-4	55.22%	52.78%	99.13%	68.88%
Exp-5	93.04%	93.04%	93.04%	93.04%
Exp-6	93.04%	92.31%	93.91%	93.10%

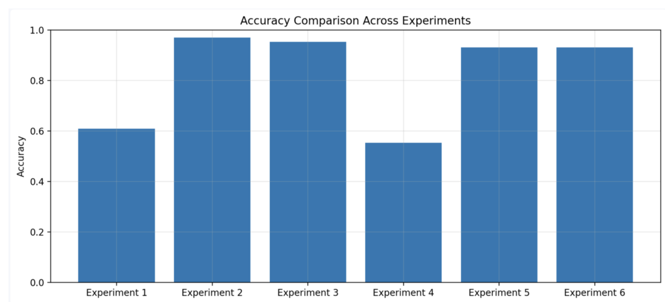

Fig. 4. Accuracy Comparison Across Different Normalization Methods

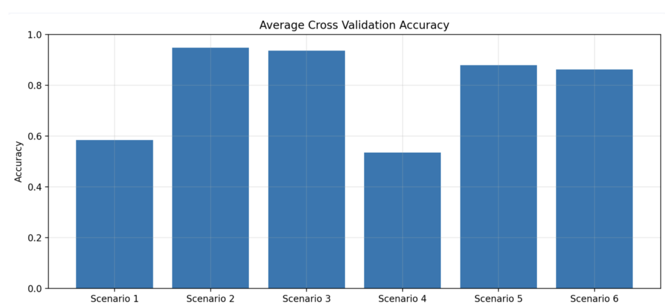

Fig. 5. Cross Validation Accuracy Comparison

Table VIII, Fig. 4, and Fig. 5 demonstrate that normalization significantly influenced the performance of the proposed DNN model for Dengue Fever classification. Experiments without normalization showed the lowest performance, particularly in Exp-1 and Exp-4, indicating that differences in feature scales negatively affected the learning process. In contrast, applying normalization methods substantially improved classification outcomes. Among all approaches, *StandardScaler* achieved the best overall results in Exp-2, obtaining 96.96% accuracy, 95.76% precision, 98.26% recall, and a 97.00% F1-Score. Meanwhile, *MinMaxScaler* also demonstrated competitive performance, especially in Exp-3, with an accuracy of 95.22%.

For the *symptoms-only dataset*, both normalization methods maintained stable classification performance above 93%, whereas experiments without normalization resulted in lower accuracy despite relatively high recall values. In addition, the *complete dataset* consistently outperformed the *symptoms-only dataset*, suggesting that laboratory examination features provided valuable clinical information for dengue classification. Overall, the findings indicate that normalization, particularly *StandardScaler*, plays an important role in improving model convergence, training stability, and overall classification performance.

C. Classification Evaluation

To further evaluate the classification capability of the proposed Deep Neural Network (DNN) model, Confusion Matrix and ROC Curve analyses were performed for each normalization method. These evaluations were conducted to analyze the model's ability to correctly classify positive and negative Dengue Fever cases as well as to measure overall classification performance.

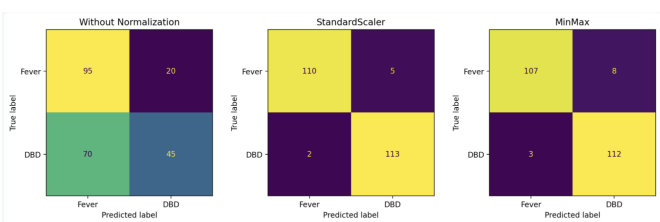

Fig. 6. Confusion Matrix Comparison for Different Normalization Methods

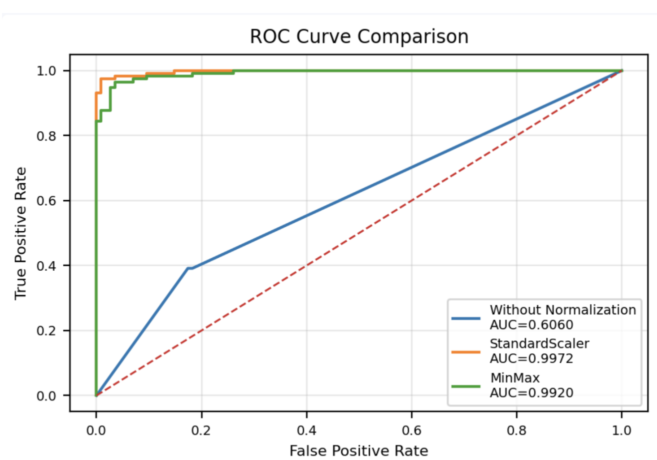

Fig. 7. ROC Curve Comparison for Different Normalization Methods

Fig. 6 and Fig. 7 present the classification performance comparison using Confusion Matrix and ROC Curve analysis for different normalization methods.

The confusion matrix results show that experiments without normalization produced a higher number of misclassifications, particularly in detecting positive Dengue Fever cases. In contrast, both *StandardScaler* and *MinMaxScaler* significantly improved prediction accuracy by reducing false classifications. Among all methods, *StandardScaler* achieved the best classification results with the highest number of correctly predicted samples.

Similarly, the ROC Curve analysis demonstrates that normalization substantially improved the classification capability of the proposed Deep Neural Network model. The model without normalization achieved a relatively low AUC value of 0.6060, while *StandardScaler* and *MinMaxScaler* achieved significantly higher AUC values of 0.9972 and 0.9920, respectively.

These findings confirm that normalization plays an important role in improving model sensitivity, specificity, and overall classification performance for Dengue Fever prediction.

V. Discussion

The experimental results demonstrate that normalization significantly influenced the performance of the proposed Deep Neural Network (DNN) model for Dengue Fever classification. Among all normalization methods, *StandardScaler* achieved the best overall performance in terms of accuracy, F1-Score, and ROC-AUC value. These findings indicate that feature scaling improves the stability of the training process and enables the model to learn feature patterns more effectively.

The results also show that the complete dataset containing both symptom and laboratory examination features consistently outperformed the symptoms-only dataset. Laboratory attributes such as platelet count, leukocyte count, hematocrit, and hemoglobin contributed important clinical information that improved the classification capability of the model. Meanwhile, the symptoms-only dataset still achieved relatively high performance, indicating its potential for early screening and preliminary self-assessment before laboratory examination is conducted.

In addition, the implementation of 5-Fold Cross Validation demonstrated that the proposed model achieved stable and consistent performance across different experimental scenarios. The relatively small gap between training and validation performance indicates that the model generalized well and experienced minimal overfitting during the training process.

Overall, the proposed DNN-based classification framework shows strong potential for supporting intelligent healthcare systems for early Dengue Fever prediction. The experimental findings also confirm that proper normalization plays an important role in improving Deep Learning classification performance on clinical datasets with heterogeneous feature scales.

VI. Conclusion

This study proposed a Deep Neural Network (DNN)-based classification framework for Dengue Fever prediction using clinical symptom and laboratory examination data. The study primarily focused on analyzing the impact of normalization methods on classification performance using different dataset configurations.

The experimental results demonstrate that normalization significantly improved the performance of the proposed model compared to experiments without normalization. Among all methods, *StandardScaler* achieved the best overall performance with the highest accuracy, F1-Score, and ROC-AUC values. In addition, the complete dataset containing laboratory examination features consistently produced better classification results than the symptoms-only dataset.

The implementation of 5-Fold Cross Validation also confirmed that the proposed model achieved stable and reliable performance with good generalization capability. Overall, the proposed framework shows promising potential for supporting early Dengue Fever prediction and healthcare decision support systems.

VII. Future Work

Future studies may improve the proposed framework by utilizing larger and more diverse clinical datasets collected from multiple healthcare institutions. Additional experiments using more advanced Deep Learning architectures such as CNN, LSTM, or hybrid models may also enhance classification performance.

Furthermore, future research can focus on integrating the proposed model into real-time healthcare applications or web-based systems to support early Dengue Fever screening and clinical decision-making processes.

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